

Experiment 9

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Subject Name: ADBMS

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Section/Group: KRG 3_A

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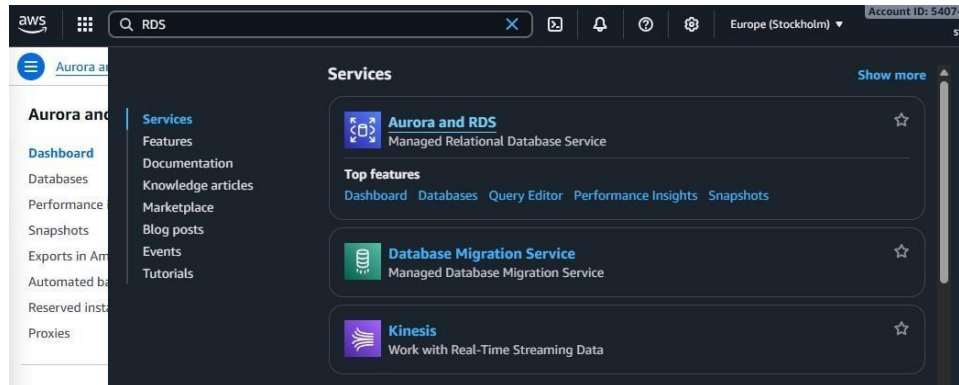
Subject Code: 23CSP-333

1. Aim:

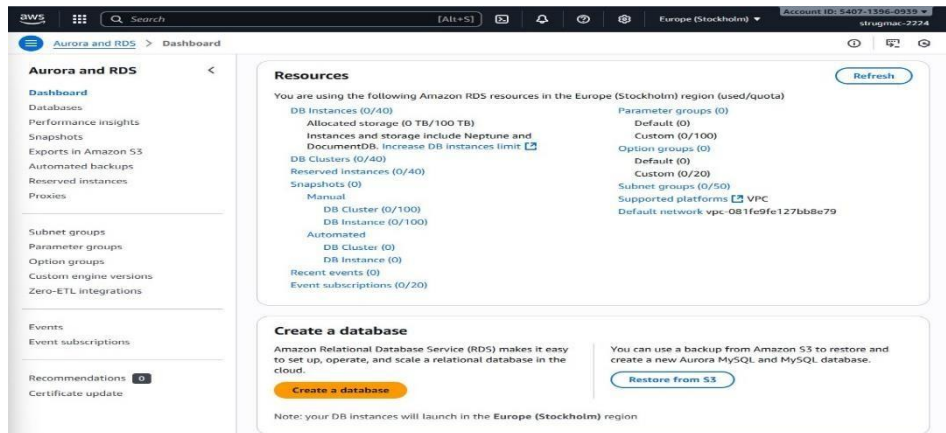
AWS RDS

- OVERVIEW OF AWS RDS
- CREATION OF DATABASE INSTANCE ON AWS RDS
- SECURITY GROUPS
- CONNECTING LOCAL PGADMIN TO CLOUD RDS

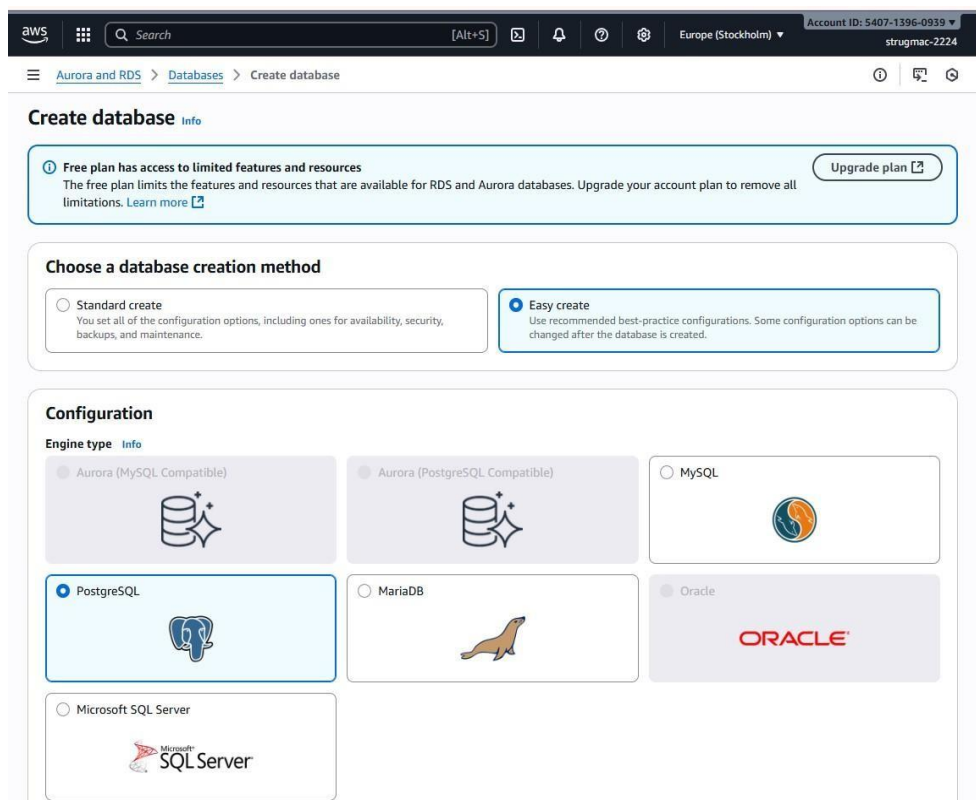
1. GO TO AWS HOMEPAGE -> CLICK ON SIGN IN-> ENTER USER NAME WITH EMAIL ADDRESS.
2. AFTER SIGN-IN -> GO TO SEARCH BAR -> SEARCH FOR RDS -> HIT ENTER



3. HOW TO CREATE MY SQL DATABASE INSTANCE ON AWS RDS?



4. CLICK ON CREATE DATABASE



5. IN THE STANDALONE CREATE, WE CAN SET EVERYTHING FOR OUR DATABASE, THE INCOMING TRAFFIC, IP ADDRESSES TO BE USED, BACKUP ETC.



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

DB instance size

☐ Production
db.r7g.xlarge
4 vCPUs
32 GiB RAM
400 GiB
1.946 USD/hour

☐ Dev/Test
db.r7g.large
2 vCPUs
16 GiB RAM
200 GiB
0.278 USD/hour

☒ Free tier
db.t4g.micro
2 vCPUs
1 GiB RAM
20 GiB
0.019 USD/hour

DB instance identifier

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

strugmac-DB

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Master username

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - **most secure**
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password or have RDS create a password that you manage.

Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password

Password strength

Very strong

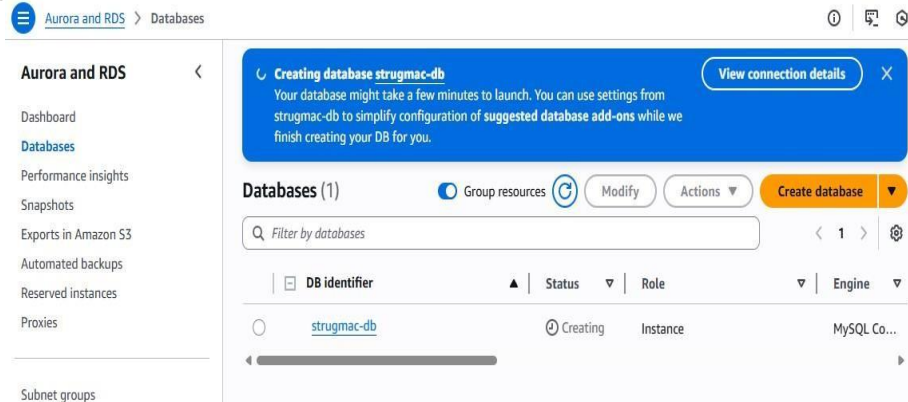
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' " @

Confirm master password

▼ View default settings for Easy create

Easy create sets the following configurations to their default values, some of which can be changed later. If you want to change any of these settings now, use Standard create.

Configuration	Value	Editable after database is created
Encryption	Enabled	No
VPC	Default VPC (vpc-081fe9fe127bb8e79)	No
Multi-AZ	No	Yes
Option group	default:mysql-8-0	Yes
Subnet group	Create new DB Subnet Group	Yes
Automatic backups	Enabled	Yes
VPC security group	default	Yes
Publicly accessible	No	Yes
Database port	3306	Yes
DB instance identifier	strugmac-DB	Yes
DB engine version	8.0.42	Yes
DB parameter group	default.mysql8.0	Yes
Monitoring type	Database Insights - Standard	Yes
Performance insights	Not enabled	Yes
Monitoring	Enabled	Yes
Maintenance	Auto minor version upgrade enabled	Yes



- Now this will create a MySQL database to me, and we want to connect to RDS for which we have to launch a server which basically will have MySQL Client installed inside it. For that we have to launch an EC2 instance,

- Launching an EC2 instance

Console Home > All services

Console Home

myApplications
[All services](#)

All services

Services by category

Compute

- EC2**
- Lightsail
- Lambda
- Batch
- Elastic Beanstalk
- Serverless Application Repository
- AWS Outposts
- EC2 Image Builder
- AWS App Runner
- AWS SimSpace Weaver
- Parallel Computing Service
- AWS Global View

Containers

Machine Learning

- Amazon SageMaker AI
- Amazon Augmented AI
- Amazon CodeGuru
- Amazon DevOps Guru
- Amazon Comprehend
- Amazon Forecast
- Amazon Fraud Detector
- Amazon Kendra
- Amazon Personalize
- Amazon Polly
- Amazon Rekognition
- Amazon Textract
- Amazon Transcribe
- Amazon Translate

EC2

Dashboard
AWS Global View
Events

▼ **Instances**

- Instances**
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations

▼ **Images**

Benefits and features

EC2 offers ultimate scalability and control

Fully resizable compute capacity to support virtually any workload. This service is best if you want:

- Highest level of control of the entire technology stack, allowing full integration with all AWS services
- Widest variety of server size options
- Widest availability of operating systems to choose from including Linux, Windows, and macOS
- Global scalability

[Find out more about EC2](#)

Launch a virtual server

[Launch instance](#)

[View dashboard](#)

[Get started walkthroughs](#)

[Get started tutorial](#)

Additional actions

[View running instances](#)

[Migrate a server](#)

EC2 > Instances

EC2

Dashboard
AWS Global View
Events

▼ **Instances**

- Instances**
- Instance Types
- Launch Templates
- Spot Requests

Instances

Find Instance by attribute or tag (case-sensitive)

All states

Name	Instance ID	Instance state	Instance type	Status check
No instances				
You do not have any instances in this region				

[Launch instances](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Firewall (security group) [Info](#)

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Network settings [Info](#)

Network [Info](#)

vpc-081fe9fe127bb8e79

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group
 ☒ Select existing security group

Common security groups [Info](#)

VPC: vpc-081fe9fe127bb8e79

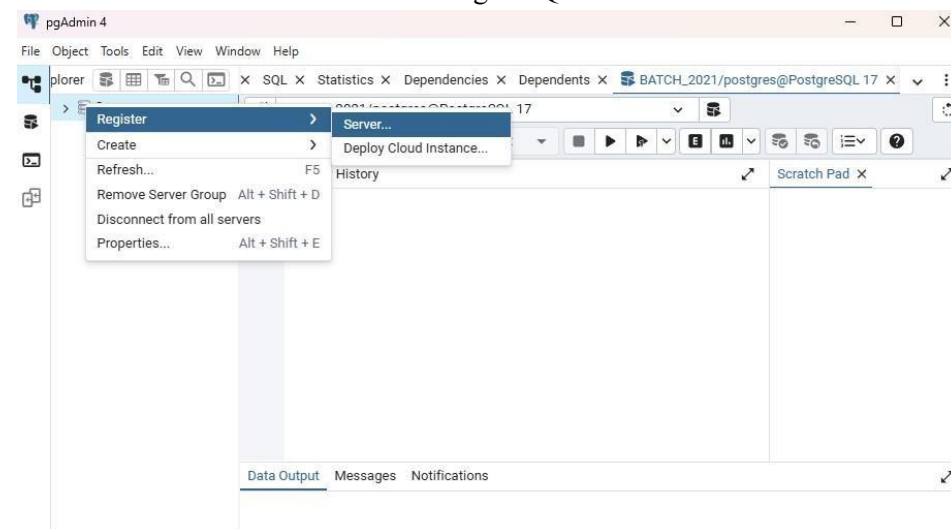
Security groups that you add or remove here will be added to or removed from all your network interfaces.

Launch instance

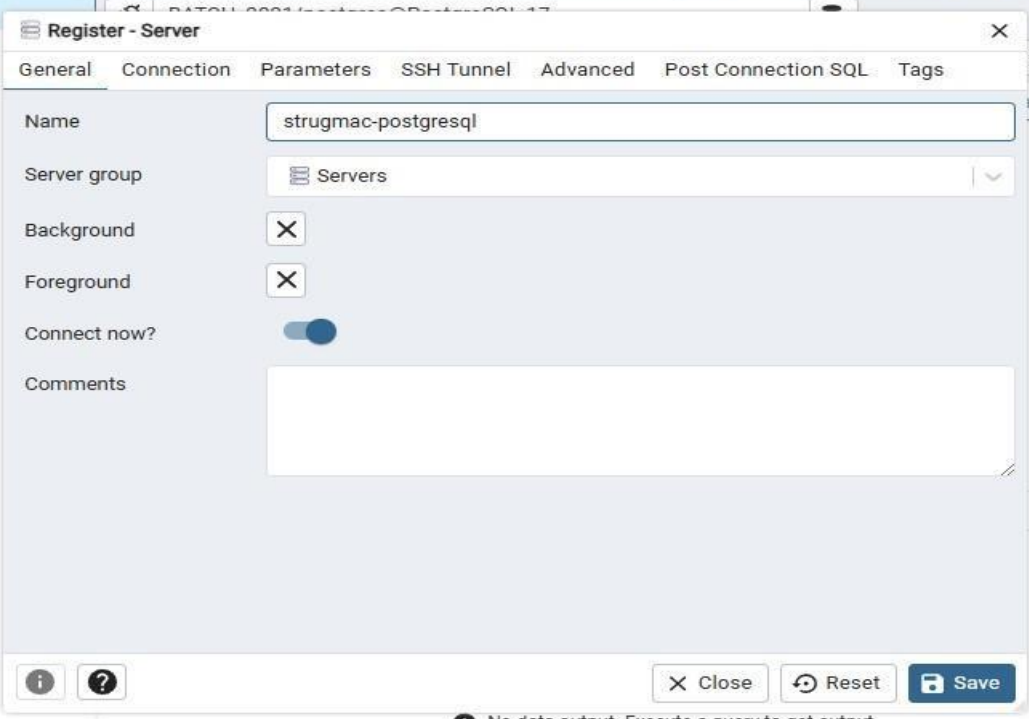
Preview code

Other option is that we can connect the Postgres AWS RDS to our local machine.

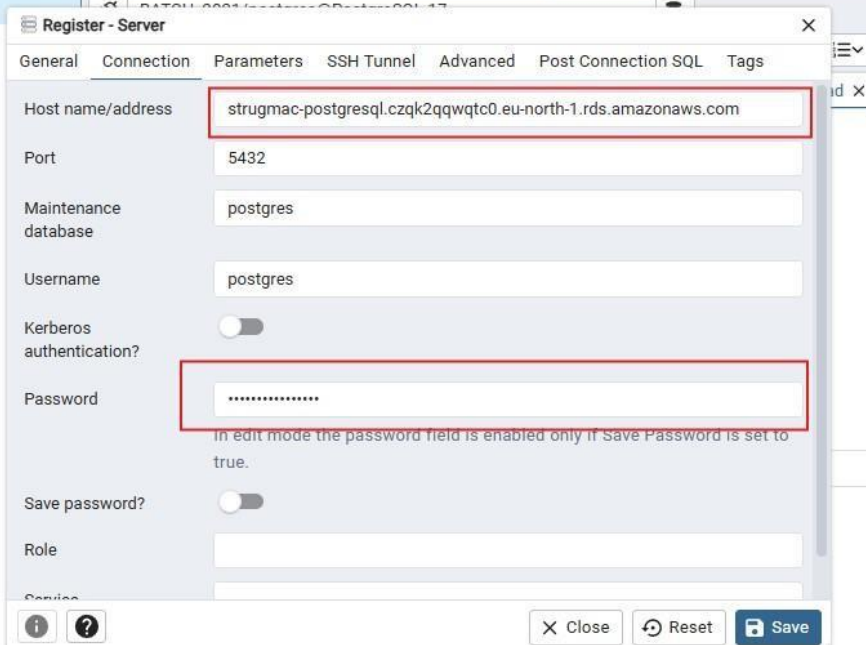
1. Create AWS RDS database for PostgreSQL 2.



2. Connect from PgAdmin.
3. Copy the API Endpoints from the dashboard of AWS RDS Database instance.

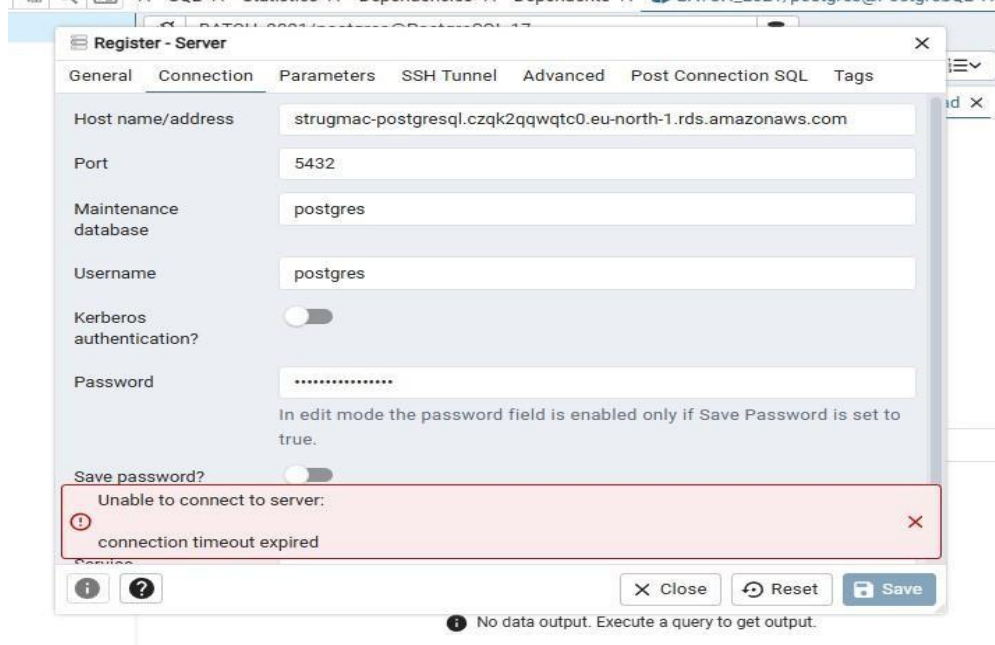


The image shows the 'Register - Server' dialog box in PgAdmin. The 'General' tab is selected. The 'Name' field contains 'strugmac-postgresql'. The 'Server group' is set to 'Servers'. The 'Background' and 'Foreground' checkboxes are unchecked. The 'Connect now?' checkbox is checked. The 'Comments' field is empty. At the bottom, there are 'Close', 'Reset', and 'Save' buttons. A status bar at the bottom indicates 'No data output. Execute a query to get output.'



The image shows the 'Register - Server' dialog box in PgAdmin, with the 'Connection' tab selected. The 'Host name/address' field contains 'strugmac-postgresql.czqk2qwwqtc0.eu-north-1.rds.amazonaws.com'. The 'Port' is '5432'. The 'Maintenance database' is 'postgres'. The 'Username' is 'postgres'. The 'Kerberos authentication?' checkbox is unchecked. The 'Password' field is masked with dots. Below the password field, a note states: 'In edit mode the password field is enabled only if Save Password is set to true.' The 'Save password?' checkbox is unchecked. The 'Role' field is empty. At the bottom, there are 'Close', 'Reset', and 'Save' buttons. A status bar at the bottom indicates 'No data output. Execute a query to get output.'

4. Click on Save



5. Might give this error as this DB instance is not available locally.

6. Change the INBOUND RULES of DB Instance from the AWS Console

